

Exploratory Test Report

webapp-auth

Report generated	2026-09-12 06:54 UTC
Run window	2026-09-12T05:40:00.630640+00:00 to 2026-09-12T06:53:21.849900+00:00
Test cases executed	20
Defects found	1
Prepared by	Exploratory Testing Planner Agent (automated)

Every outcome in this report is attributed by the test system itself. A failed run is counted as a defect only when the application was at fault; runs lost to the limits of the automated tester or to the test environment are reported separately and are not defects.

Executive Summary

A test suite of 20 automated runs was executed against the webapp-auth application, covering survey creation, survey rendering, data analysis, preprocessing, export functionality, authentication behavior, and input validation. Of the 20 runs, 1 passed, 1 revealed a defect attributable to the application, 16 were inconclusive due to automated tester limitations, and 2 were blocked by environmental conditions. The single application defect concerned the analysis module, which allowed Pearson Correlation to execute on a categorical column without the expected validation error. The most important takeaway is that the analysis module requires immediate attention regarding column type validation, while the majority of remaining tests could not reach a verdict due to navigation livelocks in the automated tester.

Scope and Method

The tests set out to verify functional behavior across survey lifecycle features, including title handling with non-Latin characters, question randomization, response export, input preservation, duplicate submission prevention, required field validation, language switching, analysis procedures, qualitative analysis, file upload size limits, quiz scoring, radio question behavior, authentication requirements, and collection status handling. The tests were executed through an automated exploratory agent that navigated the application interface, attempted to perform the specified actions, and recorded the application's responses across 445 total steps over approximately 54 minutes.

Why These Tests Were Necessary

This set of tests was selected to exercise the application's most consequential user-facing workflows, particularly those involving data integrity, survey delivery, and statistical analysis. The inclusion of Bangla character handling tests addressed localization correctness, which is central to the application's user base. The analysis and preprocessing tests targeted the correctness of statistical operations and data validation, where silent errors could lead to misleading conclusions. The authentication and collection status tests verified that access controls and survey availability behave as configured. Together, these areas represent the core value propositions of the application and the highest-risk paths for user-facing defects.

What the Tests Examined

The planned tests checked whether a survey title with Bangla characters renders correctly on a published survey, whether analysis with only one numerical column produces a message naming the missing categorical column, and whether preprocessing rejects non-numeric edits to numeric columns with a mismatch message. Additional tests checked whether a survey with shuffleQuestions enabled renders each question exactly once, whether exported responses preserve Bangla text, whether Bangla input in a text question survives save, reload, and export, and whether an unknown survey slug shows an explicit not-found message. Further tests checked duplicate submission prevention, answer retention after refused submission, whitespace-only project title rejection, project list visibility, language switch data preservation, analysis column type compatibility, qualitative analysis connection rendering, oversized file rejection, quiz score presentation, radio single-selection behavior, login-required prompting, and closed survey submission refusal.

Discussion of Findings

One defect was found and was attributable to the application. In the analysis module, the system allowed Pearson Correlation to run on the categorical column gender without blocking it, contrary to the expected behavior of presenting an error message such as Column gender is categorical, cannot run correlation. Instead, the analysis completed and displayed No significant linear correlations found, which is a silent acceptance of an invalid operation. No other application defects were identified in this run, as the remaining runs did not reach a verdict.

Limitations of This Run

Sixteen runs were terminated due to navigation livelocks in the automated tester, where repeated actions on the same page state produced no change and no verdict could be reached. These livelocks are a limitation of the testing tool and are not defects in the application. Two additional runs were blocked by environmental conditions: one because the project name field did not accept any text input, making whitespace-only validation untestable, and another because project creation repeatedly failed with Failed to create project alerts, preventing authentication behavior testing. These environmental blocks are also not defects in the application.

Recommendation

The analysis module should be reviewed to enforce column type compatibility checks before executing statistical procedures such as Pearson Correlation, and to surface a clear error message when incompatible column types are selected. The navigation livelocks affecting the majority of runs should be investigated in the automated testing tool, with adjustments to action selection and page state detection to allow tests to reach verdicts. The environmental issue preventing project creation should be examined separately, as it blocks testing of several authentication and survey configuration features. Once these obstacles are resolved, the inconclusive runs should be re-executed to complete coverage of the affected requirements.

Results by Attribution

OUTCOME	N	SHARE	MEANING
Passed	1	5%	the application behaved as expected
Defect	1	5%	a fault in the application under test
Tool limit	16	80%	our automated tester could not complete the run
Blocked	2	10%	test data, permissions, guardrail or step budget

Total execution effort: 445 steps across 3229 seconds. Of 20 runs, 2 produced evidence about the application; the remainder were lost to the tester or the environment and say nothing about its quality.

Knowledge coverage: 20 test cases recorded, 28 of 76 requirements covered, 0 user-interface states mapped.

Tests Executed

1. Survey title with non-Latin (Bangla) characters on a tester-owned published survey**Passed**

14 steps, 71s

Path: <https://webapp.example.edu/> >
<https://webapp.example.edu/?tab=projectdetails&projectId=262&source=projects> >
<https://webapp.example.edu/view-survey/610> > <https://webapp.example.edu/preview> >
<https://webapp.example.edu/view-survey/610> > <https://webapp.example.edu/preview>

2. Analysis with only one numerical column refuses with message naming missing categorical column**Tool limit**

15 steps, 51s, recorded as NAVIGATION_LIVELOCK

Path: <https://webapp.example.edu/> >
<https://webapp.example.edu/?tab=projectdetails&projectId=263&source=projects> >
<https://webapp.example.edu/?tab=analysis>

3. Preprocessing cell edit with non-numeric value on numeric column is rejected with mismatch message**Tool limit**

13 steps, 43s, recorded as NAVIGATION_LIVELOCK

Path: <https://webapp.example.edu/> >
<https://webapp.example.edu/?tab=projectdetails&projectId=278&source=projects> >
<https://webapp.example.edu/view-survey/611>

4. Survey with shuffleQuestions enabled renders every question exactly once without dropping or duplicating across renders**Tool limit**

27 steps, 96s, recorded as NAVIGATION_LIVELOCK

Path: <https://webapp.example.edu/> >
<https://webapp.example.edu/?tab=projectdetails&projectId=263&source=projects> >
<https://webapp.example.edu/view-survey/612>

5. Export responses from a Bangla survey and verify Bangla text is preserved intact in the exported file**Tool limit**

37 steps, 109s, recorded as NAVIGATION_LIVELOCK

Path: <https://webapp.example.edu/> >
<https://webapp.example.edu/?tab=projectdetails&projectId=266&source=projects> >
<https://webapp.example.edu/view-survey/596> > <https://webapp.example.edu/> >
<https://webapp.example.edu/?tab=projectdetails&projectId=266&source=projects> >
<https://webapp.example.edu/view-survey/596> > <https://webapp.example.edu/> >
<https://webapp.example.edu/?tab=projects>

6. Bangla input in a text question is preserved exactly after save, reload, and export**Tool limit**

17 steps, 324s, recorded as NAVIGATION_LIVELOCK

Path: <https://webapp.example.edu/> >
<https://webapp.example.edu/?tab=projectdetails&projectId=266&source=projects> >
<https://webapp.example.edu/view-survey/596>

7. Unknown survey slug renders explicit not-found message instead of blank page or endless spinner**Tool limit**

7 steps, 98s, recorded as NAVIGATION_LIVELOCK

Path: <https://webapp.example.edu/> > <https://webapp.example.edu/v/this-slug-does-not-exist-xyz>

8. Survey with shuffleQuestions enabled renders every question exactly once without dropping or duplicating across renders

Tool limit 55 steps, 311s, recorded as NAVIGATION_LIVELOCK

Path:	https://webapp.example.edu/	>	https://webapp.example.edu/my-templates	>
	https://webapp.example.edu/?tab=projects	>	https://webapp.example.edu/?tab=questionbank	>
	https://webapp.example.edu/?tab=projects	>	https://webapp.example.edu/?tab=questionbank	>
	https://webapp.example.edu/?tab=projects			

9. Double-click submit on survey 596 records at most one response

Tool limit 23 steps, 202s, recorded as NAVIGATION_LIVELOCK

Path:	https://webapp.example.edu/	>	https://webapp.example.edu/view-survey/596	>
	https://webapp.example.edu/preview	>	https://webapp.example.edu/view-survey/596	>
	https://webapp.example.edu/preview	>	https://webapp.example.edu/view-survey/596	>
	https://webapp.example.edu/preview			

10. Answer retention on refused submission for a required question in a tester-created survey

Tool limit 20 steps, 177s, recorded as NAVIGATION_LIVELOCK

Path: https://webapp.example.edu/

11. Attempt to create a project with a whitespace-only title and verify rejection with a visible error message

Blocked 5 steps, 48s, recorded as PRECONDITION_NOT_MET

Path: https://webapp.example.edu/

12. Project list hides another user's project after signed-in tester creates a project

Tool limit 14 steps, 79s, recorded as NAVIGATION_LIVELOCK

Path: https://webapp.example.edu/ > https://webapp.example.edu/?tab=projects

13. Language switch on a partially filled form preserves entered data and does not lose place

Tool limit 41 steps, 314s, recorded as NAVIGATION_LIVELOCK

Path:	https://webapp.example.edu/	>	https://webapp.example.edu/?tab=questionbank	>
	https://webapp.example.edu/?tab=projects			

14. Analysis refuses to run when selected column types are incompatible with the chosen procedure

Defect 11 steps, 110s, recorded as ASSERTION_FAILURE

Path: https://webapp.example.edu/ > https://webapp.example.edu/?tab=analysis

15. Verify that the qualitative analysis connection view is not rendered until at least one code and one theme are provided

Tool limit 40 steps, 202s, recorded as NAVIGATION_LIVELOCK

Path:	https://webapp.example.edu/	>	https://webapp.example.edu/qualitative-analysis	>
	https://webapp.example.edu/	>	https://webapp.example.edu/qualitative-analysis	>
	https://webapp.example.edu/	>	https://webapp.example.edu/qualitative-analysis	>
	https://webapp.example.edu/			

16. File upload question refuses an oversized file at selection time with an explicit message

Tool limit 28 steps, 558s, recorded as NAVIGATION_LIVELOCK

Path:	https://webapp.example.edu/	>	https://webapp.example.edu/?tab=questionbank	>
	https://webapp.example.edu/?tab=projects			

17. Verify that a survey with is_quiz enabled and point values set presents the score to the respondent only according to the marks-release policy

Tool limit 24 steps, 127s, recorded as NAVIGATION_LIVELOCK

Path: <https://webapp.example.edu/>

18. Radio question permits only one selection - selecting a second option replaces the first

Tool limit 21 steps, 124s, recorded as NAVIGATION_LIVELOCK

Path: <https://webapp.example.edu/> > <https://webapp.example.edu/settings> >
<https://webapp.example.edu/survey/view/596> > <https://webapp.example.edu/settings> >
<https://webapp.example.edu/> > <https://webapp.example.edu/survey/view/596> > <https://webapp.example.edu/>
 > <https://webapp.example.edu/survey/view/596>

19. Verify that a survey with isLoggedInRequired enabled shows a sign-in prompt immediately, not after filling the form

Blocked 21 steps, 112s, recorded as PRECONDITION_NOT_MET

Path: <https://webapp.example.edu/>

20. Verify that a survey with collect_response=false shows an explicit closed message and refuses submission

Tool limit 12 steps, 74s, recorded as NAVIGATION_LIVELOCK

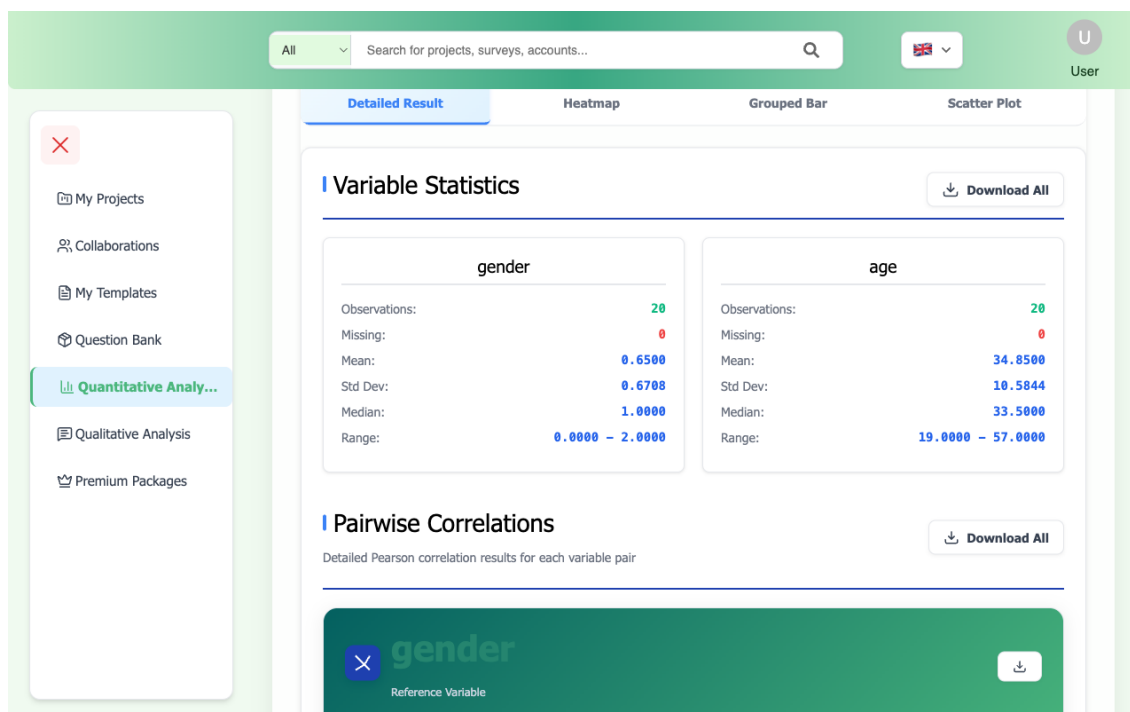
Path: <https://webapp.example.edu/> > <https://webapp.example.edu/?tab=projects>

Defects and Where They Occurred

Analysis refuses to run when selected column types are incompatible with the chosen procedure

Where: <https://webapp.example.edu/> > <https://webapp.example.edu/?tab=analysis>

Classified as: ASSERTION_FAILURE (a fault in the application)



Screen captured at the point of failure (TC-014-failed.png).

Runs That Are Not Defects

The runs below did not complete, but the cause lies with the automated tester or the test environment rather than the application. They are excluded from the defect count above and must not be read as quality problems in the software under test.

Analysis with only one numerical column refuses with message naming missing categorical column

Tool limit: NAVIGATION_LIVELOCK - our automated tester could not complete the run

Preprocessing cell edit with non-numeric value on numeric column is rejected with mismatch message

Tool limit: NAVIGATION_LIVELOCK - our automated tester could not complete the run

Survey with shuffleQuestions enabled renders every question exactly once without dropping or duplicating across renders

Tool limit: NAVIGATION_LIVELOCK - our automated tester could not complete the run

Export responses from a Bangla survey and verify Bangla text is preserved intact in the exported file

Tool limit: NAVIGATION_LIVELOCK - our automated tester could not complete the run

Bangla input in a text question is preserved exactly after save, reload, and export

Tool limit: NAVIGATION_LIVELOCK - our automated tester could not complete the run

Unknown survey slug renders explicit not-found message instead of blank page or endless spinner

Tool limit: NAVIGATION_LIVELOCK - our automated tester could not complete the run

Survey with shuffleQuestions enabled renders every question exactly once without dropping or duplicating across renders

Tool limit: NAVIGATION_LIVELOCK - our automated tester could not complete the run

Double-click submit on survey 596 records at most one response

Tool limit: NAVIGATION_LIVELOCK - our automated tester could not complete the run

Answer retention on refused submission for a required question in a tester-created survey

Tool limit: NAVIGATION_LIVELOCK - our automated tester could not complete the run

Attempt to create a project with a whitespace-only title and verify rejection with a visible error message

Blocked: PRECONDITION_NOT_MET - test data, permissions, guardrail or step budget

Project list hides another user's project after signed-in tester creates a project

Tool limit: NAVIGATION_LIVELOCK - our automated tester could not complete the run

Language switch on a partially filled form preserves entered data and does not lose place

Tool limit: NAVIGATION_LIVELOCK - our automated tester could not complete the run

Verify that the qualitative analysis connection view is not rendered until at least one code and one theme are provided

Tool limit: NAVIGATION_LIVELOCK - our automated tester could not complete the run

File upload question refuses an oversized file at selection time with an explicit message

Tool limit: NAVIGATION_LIVELOCK - our automated tester could not complete the run

Verify that a survey with is_quiz enabled and point values set presents the score to the respondent only according to the marks-release policy

Tool limit: NAVIGATION_LIVELOCK - our automated tester could not complete the run

Radio question permits only one selection - selecting a second option replaces the first

Tool limit: NAVIGATION_LIVELOCK - our automated tester could not complete the run

Verify that a survey with isLoggedInRequired enabled shows a sign-in prompt immediately, not after filling the form

Blocked: PRECONDITION_NOT_MET - test data, permissions, guardrail or step budget

Verify that a survey with collect_response=false shows an explicit closed message and refuses submission

Tool limit: NAVIGATION_LIVELOCK - our automated tester could not complete the run